CR23532

CYBER AND DIGITAL FORENSICS

CASE STUDY ASSIGNMENTS – II

Assignments 2(1), 2(2), 2(3), 2(4) and 2(5)

STUDENT DETAILS

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Prepared as a structured assignment draft. Review the examples and make any required course-specific edits before submission.

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ASSIGNMENT 2(1): Suspected Unauthorized Access to a College Server

Unique scenario title: Investigation of After-Hours Access to Confidential Examination Files

1. Introduction

A college administrator reports that confidential examination-related files were accessed and copied from a departmental server outside normal working hours. Potential evidence includes the complaint, a suspect workstation, server access logs, a USB device and relevant files. The objective is to establish what the evidence supports without contaminating the original data. All example times below are hypothetical.

2. Identification, Collection, Examination, Analysis and Presentation

Question 1: Explain the five forensic phases.

Identification: Record the complaint, authorized scope, suspected time range, systems, relevant users and available evidence. Assign evidence IDs and photograph the initial condition.

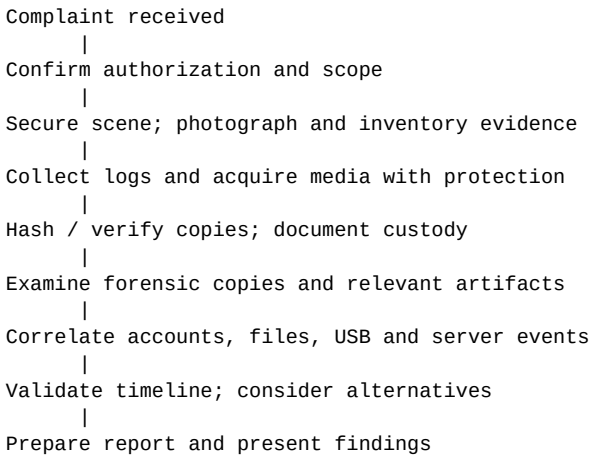
Collection: Secure devices and logs under written authorization. Record make, model, serial number, date/time, collector and condition. Use a suitable write blocker for storage acquisition where applicable, preserve logs in a verifiable export and retain originals securely.

Examination: Work on forensic images or verified copies. Examine relevant file-system metadata, login/event records, USB connection artifacts, application records and server access logs. Record tools, versions, settings, hashes and errors.

Analysis: Correlate workstation activity, server logs, USB artifacts and file metadata. Determine which account accessed which resource and whether copying is supported. Test alternative explanations and separate facts from interpretations.

Presentation: Prepare a report containing scope, methods, evidence inventory, custody, verified observations, timeline, limitations and conclusions. Explain the findings in plain language.

3. Investigation flowchart



4. Evidence table

Evidence item	Type / value	Why relevant	Caution
Administrator complaint	Documentary; contextual	Defines allegation and initial time range.	An allegation is not proof.
Suspect workstation	Digital; potentially direct/corroborating	May contain accounts, files, logs and USB traces.	Shared or compromised accounts complicate attribution.
Server access logs	Digital; potentially direct for recorded events	May show account, time, source and requested resource.	Validate completeness, clock and log meaning.
USB device	Digital; potentially corroborating	May contain copied files and device artifacts.	Presence alone does not prove who copied data.
Examination files	Target evidence	Establishes which files were of interest.	Protect confidentiality; compare hashes if appropriate.
File-system and USB artifacts	Corroborating	May support a sequence of file and device activity.	Timestamps may be incomplete or ambiguous.

5. Collection and preservation precautions

Question 4: Precautions

• Obtain proper institutional/legal authorization and keep collection within scope.

• Photograph the scene; record serial numbers, connections, condition, date/time and collector.

• Use a suitable write blocker during storage acquisition where possible; document any exception.

• Create forensic images or verified copies and record cryptographic hashes.

• Keep original evidence secured and perform examination on working copies.

• Maintain chain-of-custody entries for each transfer, opening and examination.

• Record tool versions, settings, start/end times, errors and unreadable sectors.

• Restrict access to confidential and unrelated personal information.

6. Establishing a timeline

Question 5: Timeline method

Correlate timestamps from server logs, operating-system logs, file metadata, USB connection artifacts and relevant application records. Preserve original timestamps and identify time zone, daylight-saving convention and possible clock offset. Convert to a common reference time only while retaining the original value. Correlate by account, file path, device identifier, source address and content/hash where available. A matching timestamp alone does not prove causation.

Hypothetical time	Source	Event	Interpretation
21:05	Workstation log	Account session begins	Shows a session, not necessarily the human operator.
21:12	USB artifact	USB connected	Supports device presence.
21:18	Server log	Examination file requested	Evidence of a logged request if log integrity is validated.
21:22	File/USB artifacts	Relevant file appears on USB	May support copying if corroborated.
21:31	USB artifact	Device removal recorded	Supports end of connection, not necessarily end of all activity.

7. Final report and non-technical presentation

Question 6: Report contents

• Case identification, authority, scope and questions to be answered.

• Executive summary in plain English.

• Evidence inventory, identifiers, acquisition details, hashes and chain of custody.

• Tools and versions, examination methods, artifacts and reproducible steps.

• Timeline with source, timestamp, time zone and confidence.

• Findings separated from interpretation, with limitations and alternative explanations.

• Conclusion answering the investigative questions without overstating certainty.

• Appendices with relevant screenshots, log excerpts, hash manifests and technical details.

For non-technical authorities, explain what was recorded, what the records do and do not prove, how confident the team is, and what remains unknown. Avoid unexplained jargon.

8. Legal and ethical considerations

Question 7: Legal and ethical issues

Investigators should have valid institutional and, where required, legal authorization. Collection must stay within scope, respect privacy and confidentiality, and preserve potentially exculpatory as well as incriminating evidence. Access to examination material should be restricted. The investigation should be impartial and reproducible; conclusions must distinguish facts from inference and be shared only with authorized recipients.

9. Sample forensic report structure

Section	Contents
Case details	Case ID, date, examiner, authorization and scope.
Executive summary	Allegation, work performed and main supported findings.
Evidence inventory	Item IDs, descriptions, serial numbers, custody and condition.
Acquisition and integrity	Methods, write protection, tools, hashes and errors.
Examination and analysis	Artifacts reviewed and correlation approach.
Timeline and findings	Events, source references, confidence and interpretation.
Limitations and conclusion	Alternative explanations, unresolved questions and supported conclusion.
Appendices	Hash manifests, screenshots, log excerpts and custody forms.

10. Conclusion

A defensible investigation combines server-side records with workstation and removable-media artifacts. Preservation, verified copies, timeline correlation and cautious attribution are essential. The report must state only what the evidence supports.

ASSIGNMENT 2(2): Linking a Suspicious File to a User Using File-System Evidence

Unique scenario title: Attribution Review of a Suspicious Executable on a Shared Staff Desktop

1. Introduction

A shared desktop contains a suspicious executable, hidden files, encrypted documents, browser artifacts and multiple user accounts. Investigators must assess whether evidence links a particular account to creating or accessing the files. ILookIX may support systematic examination, but its output must be validated and interpreted.

2. Hardware and software inventory

Question 8: Components to document

Record the computer make/model and serial number; CPU and RAM; internal/external storage models, capacities and serial numbers; partitions and interfaces; connected USB devices; operating-system name, version, build and architecture; local/domain accounts, profile paths and privileges; installed browsers and relevant applications; security tools; network adapters and configuration; and the forensic workstation, ILookIX version, settings and examiner identity. Record the system condition and examination date/time.

Category	Details to record	Purpose
Computer hardware	Make/model, serial, CPU, RAM, condition	Identifies the system.
Storage	Disk/USB IDs, capacity, partitions, interfaces	Supports correct acquisition.
Operating system	Name, version, build, architecture	Determines artifact locations and interpretation.
Accounts	Usernames, profile paths, privileges	Helps distinguish user profiles.
Applications	Browsers, document tools, security software	Identifies likely artifact sources.
Network	Adapters and relevant configuration/logs	May correlate online activity.
Forensic environment	ILookIX version, settings, examiner station	Supports reproducibility.

3. Storage, OS, file systems and metadata

Question 9: How these sources help

Storage devices may preserve allocated files and remnants in unallocated space. Operating-system artifacts can record accounts, logons, application use, removable-device connections and system events. File systems organize paths, directories, allocation records and metadata. Metadata may include names, sizes, timestamps, attributes and application-specific properties. Correlation across sources is stronger than relying on one filename or timestamp.

4. Hidden and encrypted files

Question 10: Effects and handling

Hidden attributes or unusual paths can cause files to be overlooked during ordinary browsing; forensic listings and searches may locate them within the authorized scope. Encryption may prevent content review without the correct key or authorized credentials. Preserve the original, record path, size, metadata and hash, and document authorized recovery attempts. Do not alter or overwrite the original. Failure to decrypt is a limitation, not proof of wrongdoing.

5. Evidence-linkage matrix

Question 11: Correlating artifacts

Artifact	Hypothetical observation	Possible link	Strength / caution
Filename/path	Executable under a user's Downloads folder	Connects file to profile location.	Supporting; software or another user may write there.
File metadata	Times near incident	Places activity in a time window.	Copying and clock settings can affect times.
Account/session logs	Account A logged in	Provides session context.	Does not prove who operated the device.
Browser download record	Matching filename or URL	May explain file origin.	Correlate time, URL, size and hash.

Artifact	Hypothetical observation	Possible link	Strength / caution
USB artifact	USB connected near file activity	Supports transfer opportunity.	Connection alone does not prove transfer.
Application/security log	Execution associated with a session	May link execution to account/session.	Validate source and event meaning.
File hash	Same hash in two locations	Supports identical content.	Does not identify creator or copier.

6. ILookIX analysis workflow

Question 12: Suggested procedure

Exact menu names depend on the installed version; follow the tool documentation and laboratory procedure. The general workflow is:

```
1. Confirm authorization, scope and evidence IDs
2. Acquire/receive the image; preserve original media
3. Verify the image using recorded hash values
4. Create a case in ILookIX and import the image
5. Select relevant volumes/file systems and index them
6. Search by filename, extension, path, hash and keywords
7. Review hidden/deleted items and available metadata
8. Inspect account, browser, log and USB artifacts
9. Export findings with source references and hashes
10. Correlate accounts/timestamps and validate manually
11. Record tool version, settings, errors and limitations
12. Report findings with reproducible references
```

This is an intended workflow, not a claim that ILookIX was actually run. Only report results supported by documented examination.

7. Direct versus contextual evidence

Question 13: Difference

A direct-linkage artifact closely records a relevant operation under an authenticated account or session—for example, a validated audit record showing a specific operation. It may be strong evidence of recorded account activity, but not necessarily the real-world identity of the operator. Contextual evidence, such as a timestamp, profile path, browser history entry or USB connection, helps explain or corroborate events but usually cannot independently identify who acted.

8. Limitations and alternative explanations

Question 14: Limitations

• The workstation or account may be shared.

• An account may have been compromised or remotely controlled.

• Clock drift, time zones or timestamp semantics may distort chronology.

• Automated applications, updates, antivirus tools or synchronization may create/change files.

• A file may have been copied from another profile or shared location.

• Logs may be incomplete, rotated, overwritten or altered.

• A matching hash proves content identity, not the identity of its creator.

9. Conclusion

Attribution should rely on multiple independent artifacts linking an account/session, a relevant operation and a consistent timeline. ILookIX can organize evidence, but investigators must validate results and report alternatives. Evidence may support an account-level link without proving who physically operated the computer.

ASSIGNMENT 2(3): Forensic Imaging and Safekeeping of a Suspect Storage Device

Unique scenario title: Verified Acquisition of a Laptop and Attached USB Drive

1. Introduction

A laptop and attached USB drive are seized during a suspected data-theft investigation. The aim is to collect and preserve the devices, create verified forensic images, use data carving where required and maintain chain of custody. IXImager and an ASB container are considered in the proposed workflow.

2. Safe collection and transport

Question 15: Step-by-step procedure

• Confirm authority and define which devices and data are in scope.

• Photograph the laptop, USB, ports, cables, screen and physical condition.

• Record make/model, serial numbers, asset labels, connections, power/network state and visible damage.

• Do not browse files or casually operate the system. If powered on, a trained examiner should decide how to handle it because shutdown can lose volatile evidence or encryption state; document every action.

• Prevent remote changes using an appropriate, documented isolation method when justified.

• Label and package laptop and USB separately in suitable evidence packaging; use tamper-evident seals.

• Record collector, date/time, location, item ID, seal number, condition and each transfer.

• Transport securely and store in an access-controlled evidence area.

3. Physical acquisition

Question 16: Purpose

Physical acquisition aims to capture a bit-for-bit representation of storage, including allocated areas and, where accessible, unallocated space and sectors. A forensic image provides a stable representation for repeatable analysis. Examining original media can change metadata or other data and reduce reproducibility, so analysis normally uses a verified image or working copy while the original remains secured. Bad sectors or inaccessible areas must be recorded.

4. IXImager imaging workflow

Question 17: Proposed procedure

Exact IXImager screens and supported output formats depend on the installed version. Follow the approved laboratory manual and document actual options selected.

```
Identify source by model / serial / capacity
                |
Photograph and document source condition
                |
Connect through suitable write blocker
                |
Select physical acquisition and destination
                |
Record tool/version, format, settings and time
                |
Acquire image; log errors and unreadable sectors
                |
Calculate and record supported cryptographic hashes
                |
Verify image using the documented method
                |
Secure original; create controlled working copy
                |
Analyze working copy and document every action
```

5. Hash integrity

Question 18: Hash values

A cryptographic hash such as SHA-256 produces a fixed-length digest of the data supplied to it. Record the acquisition hash and verify the resulting image using the method supported by the tool. Matching hashes for the same data support that the compared data are consistent at the time of calculation. Identify exactly what was hashed (source data, image file or container), algorithm, tool and time. Hash equality supports integrity, not the truth of the contents or the legality of collection.

6. ASB container

Question 19: Role and accompanying information

Where the laboratory's approved workflow supports the ASB container format, it may be used to package or manage forensic evidence and associated case information. Verify its capabilities in the relevant tool documentation instead of assuming every metadata field is embedded. Retain evidence ID, source details, acquisition time/time zone, examiner, tool/version, settings, format, hashes, errors, verification result, storage location, access history and chain-of-custody records.

7. Data carving

Question 20: Purpose and limitations

Data carving searches raw data for recognizable file signatures or structures when file-system records are missing, damaged or unavailable. It can recover complete files or fragments from unallocated space. Carving may lose the original filename, path, timestamps and ownership information; fragments may be incomplete and signature matches may be false positives. Validate recovered content, record the tool and settings, and do not infer user identity from a carved file alone.

8. Date/time problems

Question 21: Timestamp uncertainty

• Preserve the original timestamp exactly as shown by the source.

• Identify the time zone, UTC offset and daylight-saving convention where known.

• Check the system clock against a reliable reference where possible and record any offset.

• Distinguish creation, modification, access, change and application-specific times.

• Document conversion methods and uncertainty; do not silently normalize conflicting values.

• Consider timestamp changes caused by copying, restoration, synchronization or software.

9. Chain-of-custody table

Step	Custodian	Date/time	Action	Integrity / condition
1	First responder	To be recorded	Seized and labeled laptop and USB	Photographs and item IDs recorded.
2	Evidence officer	To be recorded	Sealed and transferred to secure storage	Seal number and condition recorded.
3	Forensic examiner	To be recorded	Received evidence for imaging	Transfer and seal condition documented.
4	Forensic examiner	To be recorded	Acquired and verified image	Tool, hashes and errors recorded.
5	Examiner	To be recorded	Analyzed working copy	Original media remained secured.
6	Evidence officer	To be recorded	Returned evidence to storage	Seal/storage location updated.

10. Acquisition documentation checklist

Question 22: Required records

• Source device identity, serial number, capacity and condition.

• Acquisition tool/version, write-protection method and settings.

• Image format/container, destination and acquisition start/end time.

• Source and image hash values where applicable and verification result.

• Errors, warnings, bad sectors and inaccessible areas.

• Examiner identity, evidence IDs, storage location and custody entries.

11. Conclusion

A defensible imaging process protects the original media and creates a verified working representation. Write protection, hashing, documentation, controlled storage and chain of custody are essential. Carving is useful when file-system information is unavailable, but its limitations must be reported.

ASSIGNMENT 2(4): Recovery of Deleted Evidence from Volume Shadow Copies

Unique scenario title: Reconstructing Document Activity from Windows Shadow Copies

1. Introduction

An employee is suspected of deleting documents and browser artifacts after an unauthorized activity. A forensic image of a Windows system is available, and Volume Shadow Copies may preserve earlier versions of changed or deleted items. Investigators must select relevant evidence, analyze events and validate any recovered artifact.

2. Evidence selection and categorization

Question 23: Selection method

Define the investigative question and relevant time range first. Identify potential artifacts from the forensic image and shadow copies, assign evidence IDs and preserve source references. Categorize evidence by relevance and evidential role. Filtering can reduce irrelevant material for detailed review, but the original image should remain unchanged and exclusions should be documented so potentially important evidence is not silently discarded.

3. Evidence classification

Question 24: Classification scheme

Class	Examples	Purpose / handling
User-created files	Documents, spreadsheets, images	Potential target or activity evidence; record path, metadata and hash.
System/application files	Event logs, application databases, configuration	Context and reconstruction of activity.
Potentially suspicious files	Unknown executable, unusual archive, script	Requires validation; suspicious appearance alone is not proof.
Deleted/recovered artifacts	Recovered document, deleted browser record	May reveal earlier activity; preserve recovery source and limits.
Supporting/contextual evidence	Account data, settings, network information	Helps interpret primary evidence and alternative explanations.

4. Event Analysis

Question 25: Role of Event Analysis

Event Analysis organizes artifacts into a chronological sequence. Investigators compare file events, user sessions, application activity, browser records and other timestamps to test whether a plausible sequence emerges. Each event should retain its source, timestamp, time zone and confidence. Conflicting timestamps should be investigated rather than automatically choosing one source.

5. Cloud Analysis and Lead Analysis

Question 26: Purpose

Cloud Analysis examines lawfully available cloud-stored records, synchronization events, access logs and online evidence to ask whether a file was uploaded, downloaded, shared or accessed from an account or device. Lead Analysis develops and tests investigative leads by connecting known artifacts to questions such as who, what, when, where and how. Leads are hypotheses to verify, not findings by themselves. Both must stay within the authorized scope.

6. Volume Shadow Copy analysis

Question 27: How it helps

Volume Shadow Copies can preserve earlier states of files and some system artifacts. Investigators compare an item in a snapshot with its current version to identify content or metadata changes. A file deleted or modified in the current file system may remain available in an earlier snapshot. The examiner should document the snapshot identifier/time, source path, recovery method, hashes where applicable and whether the recovered data are complete.

7. Hypothetical timeline

Question 28: Example sequence

Time	Event	Source	Interpretation
09:10	Document created	File-system metadata	Initial creation time, subject to timestamp validation.
10:25	Document modified	Metadata/content comparison	Suggests a changed version.
10:40	Document copied	Application/USB artifact	Possible transfer if independently corroborated.
11:05	Current document deleted	File-system evidence	Indicates deletion record/action if validated.
Later	Earlier version found	Volume Shadow Copy	Historical version potentially recoverable.
Later	Recovered item validated	Hash/content/source comparison	May support reconstruction; does not alone identify who deleted it.

8. Validation checklist

Question 29: Validate recovered artifacts

• Confirm the artifact came from the forensic image or a documented shadow-copy snapshot.

• Record the exact snapshot reference and original path.

• Calculate and record hashes where appropriate; compare versions and content.

• Correlate with independent artifacts such as logs, browser records or account activity.

• Check timestamp source, time zone and system-clock assumptions.

• Consider whether applications or synchronization could have created the artifact.

• Document missing data, partial recovery, errors and alternative explanations.

• Include the item as a finding only if evidence supports the stated interpretation.

9. Errors and alternative explanations

Question 30: Possible sources of error

• Incorrect or drifting system clock, time-zone differences or timestamp conventions.

• Automatic operating-system or application activity.

• Synchronization or backup processes changing file states.

• Copying that preserves or changes timestamps.

• Partial recovery, damaged data or misleading file signatures.

• Shadow-copy creation before or after the event of interest.

• Deletion may have been performed by an automated process or another account.

10. Conclusion

Volume Shadow Copy analysis can provide historical context when current evidence has been changed or deleted. Reliable findings require classification, event correlation, validation and documentation of timestamp uncertainty. Recovery alone does not prove who performed the original action; attribution requires corroboration.

ASSIGNMENT 2(5): Multi-Source Investigation of a Suspected Cyber Incident

Unique scenario title: Cross-Device Correlation of an Unauthorized Online-Service Access Event

1. Introduction

A student account is suspected of being used in a cyber incident involving unauthorized access to an online service. Subject to proper authorization, investigators may examine a Windows computer, browser artifacts, email and messaging records, a mobile phone and Internet/cloud records. The aim is to correlate independent evidence while accounting for the limitations of each source.

2. Windows and other operating-system evidence

Question 31: Potential evidence

• User accounts, login/session artifacts and relevant system event logs.

• File-system metadata, relevant documents and application artifacts.

• Browser databases, download records and cached files where available.

• Removable-device connection artifacts and network configuration/logs.

• Installed applications, configuration data and security alerts.

• Other operating systems may contain analogous account, file-system, application and event records; exact locations and meaning vary by platform.

3. Browser artifacts

Question 32: What browser evidence may reveal

Browser history may show visited URLs; download records may show file names, paths and times; cache may contain resources; cookies and session data may reflect web sessions; autofill or saved-session data may contain account-related information, subject to availability and authorization. Browser evidence must be interpreted carefully because shared devices, automated requests, synchronization, extensions, background tabs and other users can generate artifacts. A history entry does not necessarily prove that the user intentionally viewed or acted on the page.

4. Email and messaging evidence

Question 33: Types and correlation

Source	Potential evidence	Correlation method
Email	Headers, sender/recipient, subject, timestamps, attachments	Compare header times, account logs, downloaded files and attachment hashes.
Messaging	Messages, participants, timestamps, shared media	Compare with mobile artifacts and relevant online-service events.
Attachments	Downloaded or shared documents, images, archives	Compare hashes, names, paths and timestamps across devices.
Account records	Login/access events where available	Compare account activity with computer, mobile and network records.

5. Mobile phone forensics

Question 34: Considerations and challenges

Obtain appropriate authorization and document the phone's make/model, serial or device identifier, OS version, state, SIM and relevant accessories. Use an appropriate validated acquisition method and preserve the original data. Potential evidence includes calls, messages, application data, media, account information and device timestamps. Challenges include encryption, screen locks, app-specific storage, changing data, cloud synchronization, operating-system differences, limited acquisition methods and data loss caused by mishandling. Record exactly what was and was not acquired.

6. Internet and cloud evidence

Question 35: Relevant sources

Potential sources include online-service access logs, cloud-storage metadata, provider account/login records, synchronization history, IP-related records, email-provider logs, DNS or network records and application logs, where lawfully available. These records may help identify account access, file sharing, download/upload activity and source connections. Availability depends on provider retention, jurisdiction, account settings and the scope of authorization. Preserve provider exports with acquisition details and integrity information.

7. Cross-source correlation matrix

Question 36: Hypothetical artifacts

Artifact / event	Computer	Browser	Email/message	Mobile	Cloud	Interpretation
1. Account login near incident	OS session record	Service page visited		App session	Provider login record	Corroborates account activity if times and identifiers align.
2. Downloaded attachment	File in Downloads	Download record	Email attachment	Message copy possibly present	Cloud attachment record	Compare content hashes and timestamps to test file relationship.
3. Message before incident			Message timestamp and sender	Same conversation on phone	Backup/sync record if available	May establish communication sequence, not intent by itself.
4. Online-service access	Relevant local artifacts	URL/session-related artifact		Mobile app activity	Service access log	Correlate account, source address and normalized time.
5. Shared file	Local file and metadata	Download history	Shared attachment/link	File or message copy	Cloud file metadata	Matching hashes and IDs can support a common file across sources.

All rows are hypothetical examples of how evidence could be correlated; they are not claims that these artifacts were found in a real investigation.

8. Major challenges in multi-source forensics

Question 37: Challenges

• Different timestamp formats, time zones and device clock errors.

• Encryption and locked devices that restrict acquisition.

• Cloud data held by third-party providers and possibly in different jurisdictions.

• Volatile or changing online records and short retention periods.

• Different operating systems, application formats and artifact structures.

• Attribution problems when accounts or devices are shared or compromised.

• Synchronization may replicate the same artifact across several sources, so apparent agreement may not be independent corroboration.

• Large data volumes make filtering, indexing and correlation difficult.

• Provider retention policies may remove historical records.

• Legal authorization and jurisdiction may restrict access to some evidence.

• Acquisition tools may not recover all app data or may interpret formats differently.

9. Final investigative conclusion

Question 38: Distinguish evidence categories

Direct evidence: A validated service audit record documenting a relevant action under a particular account, or another reliable record closely tied to the event. It directly supports the recorded action but may not identify the human operator.

Corroborating evidence: Independent computer, browser, email, mobile or cloud artifacts that align in time and identifiers and support the same sequence. The examiner should check whether sources are truly independent or merely synchronized copies.

Unresolved questions: Who physically operated the device; whether credentials were compromised; missing provider records; uncertain timestamps; encrypted content unavailable for examination; and whether a recorded request was intentional

or automated.

10. Conclusion

A multi-source investigation should not treat one device artifact as conclusive. Strong findings arise when independent sources agree on identity, time and activity. Investigators must document direct evidence, corroboration, limitations and unresolved questions, and ensure all collection and examination are authorized.

FINAL SUBMISSION CHECKLIST

■ Fill in your name and register number on the cover page.

■ Review every hypothetical example and adjust it only if your own scenario or evidence differs.

■ Add screenshots only if you actually performed the relevant tool operation; do not claim ILookIX or IXImager was run unless you have supporting evidence.

■ Add your course-specific references or prescribed textbook references if required by your faculty.

■ Check the assignment instructions and formatting requirements before submission.

■ Keep all original evidence and any real personal data confidential.

Note: This document is a structured draft based on the case-study questions supplied. Hypothetical timelines and artifact examples are clearly illustrative; replace or supplement them with your own verified work where required.